MAHATMA GANDHI UNIVERSITY, KOTTAYAM

FIRST SEMESTER MGU-UGP (HONOURS) REGULAR
EXAMINATION NOVEMBER 2024

Multi-Disciplinary Course- **MG1MDCPES100-BASIC FIRST AID AND CPR**

(2024 ADMISSION ONWARDS)

QP CODE: 24900040 Duration: 1-hour Maximum marks: 35

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create(C), Interest (I), Appreciation (Ap) and Skill(S)

Part A **Multiple choice questions. Answer all questions. Each question carries 1 mark**

1. Which of these is a key component of first aid? (K) CO 1
   Answer: B
2. What does "informed consent" mean in the context of first aid? (K) CO 1
   Answer: C
3. Which type of joint allows for the greatest range of motion? (K) CO1,2
   Answer: B
4. When performing CPR on an infant, how many fingers should you use for chest compression. (U) CO
   Answer: B
5. In what situation would you need to wear eye protection? (K) CO 1,2
   Answer: B
6. What is the first thing you should do if you see someone collapse? (U) CO 3
   Answer: A, B, C (can give mark for writing A/B/C)
7. Which of the following is NOT a sign of a fracture? (U) CO 4
   Answer: C
8. The injury that occurs when a muscle is overstretched or torn is called …….. (K) CO 5
   Answer: B
9. Seizure is a …… (U) 5
   Answer: B
10. Which component of blood helps fight infections? (K) CO 2
    Answer: C

(10x1=10)

Part B - Short answer type questions. Answer any five questions. Each questions carries 2 marks.

11. What are the main purpose of first aid? (U) CO 1
    First aid is the immediate assistance given to a person who is ill or injured. The main purpose of first aid is to provide timely and effective assistance to individuals in emergencies. This involves preserving life, preventing further injury or illness, and promoting recovery until more advanced medical care is available.

12. Differentiate between tendons and ligaments. (U) CO 1,2

Tendons and ligaments are connective tissue, but they have distinct structures, functions, and locations in the body. **Tendon**: Tissue that connects muscles to bones and transmit the force generated by muscles to move the bones.
Ligament: Tissue that connects bones to other bones and stabilizes joints by preventing excessive or abnormal movements.

13. If a casualty is not breathing, what action should you take immediately? (A) CO 1,2

If a casualty is not breathing, immediate action should be taken as follows, gently shake the casualty and shout to see if they respond. If there is no response, immediately call for emergency services or ask someone nearby to do so. To open the airway, Tilt the casualty's head back and lift their chin. Check for breathing. Look, listen, and feel for normal breathing for no more than 10 seconds. If the casualty is not breathing, start CPR immediately.

Place your hands in the centre of their chest and begin chest compressions. Press hard and fast, about 100–120 compressions per minute, at a depth of 2 inches for adults. After 30 compressions, give 2 rescue breaths. Pinch the nose, cover their mouth, and breathe in. Continue CPR until emergency help arrives, the casualty starts breathing, or you are physically unable to continue.

14. What is the difference between adult and child CPR. (U) CO 2

The main difference between adult and child CPR lies in the technique and sequence of actions. For a child, use one of your hands. Place the heel of the hand at their sternum, which is in the center of the chest, between and slightly below their nipples. Press down hard and fast around 1.5 inches deep, or one-third the depth of the chest. For an infant, use two fingers. Place your fingers in the center of their chest, between and slightly below the nipples. For adults, place the heel of one hand in the middle of the chest. Place your other hand on top of the first hand and interlace the fingers. For adults, compress the chest to a depth of at least 2 inches. Rescue breaths follow the same ratio of 30 compressions to 2 breaths for both adults and children

15. What should you do if someone is in a drowning incident? (U) CO 2

Move the person out of the water. Check the level of response, open the airway and check breathing. If the casualty is unresponsive and not breathing normally shout for help and immediately star CPR. Give five initial rescue breaths, follow this with 30 chest compressions, then two rescue breaths. Repeat the procedure till casualty shows signs of becoming responsive or until help arrives.

16. Why should the bleeding injury elevate above the heart level? (K) CO 5

Elevating a bleeding injury above heart level is a first-aid technique used to help control bleeding. By elevating the injured area above the heart, gravity reduces the blood pressure in the blood vessels at the injury site, making it harder for blood to flow out of the wound. This reduces the

stress on the heart. Slower blood flow allows the natural clotting. Elevation also helps to reduce swelling in the injured area.

17. Explain the RICE method. (U) CO

5R.I.C.E. stands for rest, ice, compression, and elevation, and taking these simple steps following a strain, sprain, or other similar injury can help more quickly recover and get back to everyday activities.

Rest : Avoid using the injured area

Ice : Apply ice or a cold pack to the area for 20 minutes at a time every 4 hours

Compression : Wrap the area with a bandage to reduce swelling and provide support
Elevation : Keep the injured area raised above the level of the heart

(5x2=10)

Part C - Essay type questions. Answer any three questions. Each question carries 5 marks.

18. Describe the main objectives of first aid. How do these objectives help in save lives. (K) CO 1

First aid is the act of providing immediate assistance to someone who is injured or suddenly becomes ill. The goal is to support recovery and keep the situation from getting worse until professional medical help is available. The main objectives of first aid are often summarized as the "Three Ps."
Preserve Life: The main purpose of first aid is to prevent the condition from worsening. This could include performing life-saving techniques such as CPR.
Prevent Further Injury/Illness: This includes steps to avoid further injuries or conditions, such as stabilizing a person's condition or preventing an infection.
Promote Recovery: After immediate threats are managed, first aiders can help begin the recovery process by ensuring the person is comfortable and secure until further help arrives

19. What is the difference between voluntary and involuntary muscles? (K) CO 2

Muscles in the human body can be classified into voluntary and involuntary muscles based on whether their movement is under conscious control. Voluntary muscles are under our conscious control and are responsible for body movements like walking, lifting, and running. Skeletal Muscles are voluntary muscles and striated in appearance due to organized muscle fibers.
Involuntary muscles are not under our conscious control and contract and relax automatically. They are also known as unstriated or smooth muscles. Smooth Muscles are found in the walls of internal organs (e.g., stomach, intestines, and blood vessels). Cardiac Muscles are found only in the heart. Involuntary, works automatically to maintain the heartbeat.

20. Discuss the role of CPR in remote or rural areas where emergency services may be delayed. (K) CO 2

In rural or remote areas, where medical help may take longer to arrive, CPR is an essential life-saving skill. It ensures vital functions are maintained, increases survival rates, and empowers local communities to act during emergencies. Training and awareness in CPR are particularly vital in these regions to prepare for cardiac or respiratory emergencies effectively. By raising awareness, providing training opportunities, and leveraging technology, rural communities can empower residents to respond effectively to cardiac arrest situations and significantly improve the chances of survival.

21. Explain the steps to control bleeding in a first aid situation? (U) CO 4

Controlling bleeding is a crucial first-aid skill that can save lives. Before assisting, ensure the scene is safe to avoid putting yourself at risk (e.g., sharp objects, traffic). Protecting yourself prevents further harm or contamination. Do not remove anything is embedded in a wound. Apply direct pressure to the wound with your fingers or the palm of your hand over a sterile wound dressing to control the bleeding. Direct pressure helps compress blood vessels and reduces blood flow, helps clotting. Raise the injured body part above heart level. Once bleeding slows, secure the dressing in place with a bandage. If bleeding soaks through, don't remove the dressing, add another layer. Severe blood loss can cause shock, which is life-threatening Look for symptoms like pale, clammy skin, rapid breathing, or confusion. Lay the person down, keep them warm, and raise their legs if no injury prevents it.

22. You are cooking and accidently spill boiling water on your hand. Explain the first aid protocols you should take immediately after the scald? (K) CO 5

Immediately remove the affected hand from the hot water and take off any jewellery, rings, or watches near the burn area. Hold the burned area under cool (not cold) running water for 10-20 minutes, or immerse it in cool water. Do not apply ice or ice-cold water directly to the burn. Cold can damage the skin and worsen the burn. Gently pat the area dry and cover it with a clean, non-stick, sterile dressing or a clean cloth. Check the burn over the next few hours or days for signs of infection, such as increased redness, swelling, pus, or worsening pain. Infections can lead to complications and require medical attention.

(3x5=15)

…….